

# AI DEVELOPMENT PLATFORM

Comprehensive functional blueprint for a South African land, layout, design, estate, climate, costing, and build-intelligence platform

## **Purpose**

This document defines how the platform should work, what must be integrated, how the house-layout engine should think, how estate-level and stand-level analysis should interact, and what outputs are required to make the tool useful to real developers.

*Prepared for concept development and platform scoping*

## 1. Executive Summary

This platform is intended to be an all-in-one decision system for South African developers. It should ingest land information, planning and zoning rules, contour and survey data, climate conditions, estate-level information, and a detailed house brief, then convert those inputs into practical house layouts, stand-by-stand design options, estate intelligence, cost guidance, green recommendations, and bill-of-quantities outputs.

The most important engine in the whole platform is the house-layout engine. If the house is not positioned and designed properly for the exact stand, the estate logic, financial outputs, solar advice, and construction recommendations will all be weakened. For that reason, the platform must treat the layout engine as the core product, with every other module feeding into it.

### Core product promise

Give the user the most practical, buildable, efficient, visually coherent, site-aware house concept for a specific stand, while also allowing estate-wide analysis and repeatable house-type logic for larger developments.

## 2. Strategic Objective

The platform should answer three practical questions.

1. What is the smartest thing to build on this land?
2. How should the building sit on this exact stand to work with contours, sun, wind, access, privacy, and cost?
3. How can the final concept be built more efficiently, more sustainably, and with better commercial logic?

## 3. Design Principles

- Practical before flashy. The system should avoid beautiful but inefficient concepts.
- Stand-specific before generic. A design must fit the exact erf, not a theoretical template.
- Contour-aware. The system must work with the land and reduce unnecessary cut, fill, retaining, and awkward levels.
- Climate-aware. Sun, shade, wind, rainfall, local temperature patterns, and outdoor living comfort must influence design.
- Developer-friendly. Outputs must help with decision-making, not only visualisation.
- User-friendly. Complex intelligence should sit in the background while the interface remains guided and simple.
- Flexible. A user must be able to work in quick concept mode, detailed house mode, or estate mode.
- South Africa-specific. Zoning, costing logic, climate assumptions, utility options, and compliance workflows must align with local realities.

## 4. User Types

| User type                    | Primary need                                               | Typical use case                                                         |
|------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------|
| Developer                    | Feasibility, layout, estate strategy, yield and cost logic | Testing new land, optimizing an estate, choosing house types per stand   |
| Architect / concept designer | Quick but site-aware concept generation                    | Generating layout options and exterior directions before detailed design |

|                      |                                     |                                                                           |
|----------------------|-------------------------------------|---------------------------------------------------------------------------|
| Builder / QS         | Build logic and quantity extraction | Testing structural simplicity, cost range, and preliminary BOQ            |
| Landowner / investor | Visual decision-making              | Understanding what can fit on a stand and what it may look like           |
| Estate planner       | Masterplan-level logic              | Comparing stand categories, security solutions, and house-type deployment |

## 5. Platform Operating Modes

| Mode                   | Purpose                                                    | Main output                                                                                             |
|------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Single Erf Mode        | Analyse one stand and design one house for it              | Stand analysis, layout options, renders, cost estimate, BOQ draft                                       |
| Estate Masterplan Mode | Analyse the entire development first                       | Stand classification, estate opportunities, security and entrance strategy, common-area recommendations |
| Estate Stand Mode      | Design a house per stand inside an already-analysed estate | Stand-specific house concepts informed by estate rules and neighbouring conditions                      |

## 6. Full Workflow Overview

4. Create a project and choose the operating mode.
5. Upload base land and document inputs.
6. Generate a site intelligence model and, where relevant, an estate intelligence model.
7. Interpret zoning and other planning constraints into practical design rules.
8. Build a stand-specific buildable envelope and identify risk zones.
9. Complete a structured house brief with both spatial and relational requirements.
10. Generate layout options that are positioned on the stand, not floating independently.
11. Generate exterior massing and render views in the chosen style and feel.
12. Run climate, solar, wind, water, and outdoor-living optimisation.
13. Run structural simplicity and build-efficiency checks.
14. Generate cost and quantity outputs.
15. Compare options, refine the brief, and export the selected concept.

## 7. Required Inputs and Integrations

### 7.1 Core land and geometry inputs

| Input                                         | Why it matters                                             | Required capability                                           |
|-----------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------|
| Erf number                                    | Entry point for land-specific data                         | User enters erf number and platform creates the project shell |
| Google Maps / Google Earth pin or coordinates | Allows land location and orientation                       | Map-based selection and coordinate storage                    |
| Surveyor points / KML / KMZ / DWG / DXF / PDF | Defines exact stand geometry and surveyed reference points | File upload, parsing, geometry conversion                     |
| SG diagram                                    | Boundary and legal geometry reference                      | Upload and document parsing                                   |
| LG card / contour information                 | Defines slope and topography                               | Contour extraction and terrain modelling                      |

## 7.2 Legal and planning inputs

| Input                                   | Why it matters                                                                           | Required capability                               |
|-----------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------|
| Zoning certificate                      | Controls allowable use, density, coverage, FAR, height, building lines, parking and more | Document OCR/parsing and rule extraction          |
| Title deed restrictions                 | May override or add conditions                                                           | Document comparison and conflict flagging         |
| Estate design guidelines                | Controls style, materials, roof form, façade requirements and estate harmony             | Rules ingestion and downstream design constraints |
| Municipal comments / previous approvals | Useful for practical risk awareness                                                      | Document upload and notes extraction              |
| Servitude plans / overlays              | May reduce buildable area or require special treatment                                   | Spatial overlay support                           |

## 7.3 Estate-level inputs

| Input                                             | Why it matters                                                   | Required capability             |
|---------------------------------------------------|------------------------------------------------------------------|---------------------------------|
| Site Development Plan (SDP)                       | Estate-wide geometry, roads, stands, open spaces and layout      | Import and layer parsing        |
| Stand schedule                                    | Stand numbers, areas, categories, intended product               | Spreadsheet/document import     |
| Road and access layout                            | Influences entry, garage placement, traffic flow and streetscape | Spatial overlay                 |
| Security brief                                    | Needed to compare access control strategies                      | Questionnaire plus estate rules |
| Engineering and services concept, where available | Useful for drainage, water and practical build logic             | Optional upload and annotation  |

## 7.4 Environmental and climate inputs

| Input                                    | Why it matters                                                                  | Required capability                 |
|------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------|
| Location-based climate data              | Needed for wind, rainfall, sun path, temperature tendencies and outdoor comfort | Geolocated climate feed integration |
| Solar path by location                   | Drives solar positioning and room orientation                                   | Sun-path calculations               |
| Rainfall and runoff tendencies           | Influences water harvesting and drainage suggestions                            | Hydrology-aware recommendations     |
| Coastal / inland condition indicators    | Impacts corrosion, wind, humidity and envelope design                           | Context profile per site            |
| Neighbouring context and view directions | Affects privacy, windows and outdoor placement                                  | Map and/or manual user annotation   |

# 8. Core Intelligence Layers

## 8.1 Site Intelligence Layer

Builds a digital understanding of the land itself: boundaries, terrain, slope zones, access points, north, views, neighbouring constraints, and likely buildable platforms.

### 8.2 Legal Intelligence Layer

Interprets zoning, title conditions, estate rules, building lines, height limits, density, coverage, parking rules, use rights, and servitudes, then translates those into practical design constraints.

### 8.3 Climate and Natural Elements Layer

Evaluates sun, shade, seasonal light, wind patterns, rainfall, runoff, outdoor comfort, braai conditions, exposed façades, and green opportunities such as solar and water harvesting.

### 8.4 Layout Intelligence Layer

Generates the actual house logic: positioning, levels, room sequencing, circulation, privacy, views, indoor-outdoor flow, and the relationship between plan efficiency and external form.

### 8.5 Build and Cost Intelligence Layer

Assesses structural simplicity, earthworks, retaining risk, rough construction method suitability, cost range, quantity take-off logic, and greener alternatives.

## 9. The House-Layout Engine: Core Product Specification

This is the most important engine in the platform. It must not merely arrange rooms. It must take the stand, contours, climate, access, legal constraints, and house brief, then create a house that is both practical and visually convincing.

#### Non-negotiable requirement

The platform must design the house for the exact stand. It may never produce a generic plan and simply place it on the site afterwards.

### 9.1 What the layout engine must consider simultaneously

- Stand dimensions and shape
- Contours and slope direction
- Possible cut and fill implications
- Street level and approach to the erf
- Garage access practicality
- North orientation and sunlight quality
- Prevailing wind and sheltered outdoor areas
- Neighbouring stands and privacy pressure
- Views worth capturing
- Setbacks and buildable envelope
- Estate design rules
- Room relationships, split between levels, and circulation requirements
- Construction practicality and cost discipline

### 9.2 Layout engine objectives

- Create the most practical internal layout for the stand

- Reduce wasted space, especially in passages and awkward corners
- Place the house where it performs best on the stand
- Use levels intelligently on sloping land
- Create good indoor-outdoor living
- Support strong exterior composition and façade quality
- Keep the concept buildable and cost-aware
- Allow comparison between multiple options

### 9.3 Required output types from the layout engine

| Output              | Minimum standard                                                                                                                    |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Floor plan          | Proper room-by-room plan with labels, dimensions, doors, windows, stairs and logical furniture placement where useful               |
| Site plan           | House accurately positioned on stand with driveway, patios, garden space, water tank zones, solar zones and major external elements |
| 3D massing          | Terrain-aware massing model showing levels, roofs and how the building meets the slope                                              |
| Exterior renders    | At least front, street and main garden/patio view in the chosen style                                                               |
| Design rationale    | Clear explanation of why rooms and levels were positioned as they were                                                              |
| Performance overlay | Sun, wind, privacy, and build-complexity notes per option                                                                           |

## 10. Structured House Brief Builder

The house brief should combine guided dropdown selections with optional exact dimensions and a free-text layer. The platform should support both quick concept mode and detailed planning mode.

### 10.1 Brief categories

| Category               | Examples of required inputs                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------|
| General building type  | Single storey, double storey, split level, or let AI recommend                                     |
| Room counts            | Bedrooms, bathrooms, living areas, study, guest toilet, garage size                                |
| Level allocation       | Example: two bedrooms upstairs and one downstairs; living areas downstairs; pajama lounge upstairs |
| Room hierarchy         | Main bedroom larger, guest bedroom more private, children grouped together                         |
| Relationship rules     | Kitchen connected to garage; patio linked to living room; guest WC hidden from main view           |
| Dimensions and scale   | Compact, standard, generous, or exact room dimensions                                              |
| Lifestyle requirements | Braai outside, covered patio, larger living room, indoor fireplace, scullery, walk-in closet       |
| Priorities             | Minimise earthworks, maximise north sun, maximise views, keep cost lower, allow future extension   |
| Style and feel         | Modern, farmhouse, warm and natural, minimal, estate-specific look and materials                   |

### 10.2 Detailed room logic that must be supported

- The user must be able to say exactly how rooms are distributed across floors.
- The user must be able to specify that the main bedroom is larger, more private, north-facing, and linked to a certain bathroom type.
- The user must be able to request a large or standard living room, or enter exact dimensions.
- The user must be able to tell the engine which rooms must connect directly and which rooms must be separated.
- The user must be able to say when a room should open to the garden, patio, or courtyard.
- The user must be able to rank practical priorities so the AI knows what to optimise first.

### 10.3 Example brief fields

| Field group | Examples                                                                                              |
|-------------|-------------------------------------------------------------------------------------------------------|
| Bedrooms    | Total number; number upstairs; number downstairs; main bedroom requirements; guest suite requirements |
| Bathrooms   | Ensuite or shared; bath, shower, double vanity; guest toilet                                          |

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| Living spaces      | Open-plan or separate; double-volume or standard; TV lounge; dining linkage             |
| Kitchen            | Open, semi-open, closed; island; scullery; pantry; laundry connection                   |
| Garage and parking | Double garage; storage nook; direct access; visitor bays                                |
| Outdoor            | Covered patio; wind-sheltered braai; pool; drying yard; water tank placement preference |
| Build efficiency   | Simpler roof; reduce retaining; stack wet areas; easier plumbing; future extension      |
| Look and feel      | Style family; material palette; roof language; façade character                         |

## 11. Site Analysis Engine

The site analysis engine is the first technical stage after data upload. It creates the conditions the layout engine must obey and exploit.

### 11.1 Required functions

- Generate the exact stand geometry
- Create a terrain model from contour information
- Identify slope direction, steep zones and likely flatter building platforms
- Determine north and solar exposure by aspect
- Identify likely access and driveway approach practicality
- Map views, privacy edges, and neighbouring pressure zones where possible
- Show potential areas for garden, outdoor living and service zones

### 11.2 Deliverables from site analysis

- 3D site visual of the stand
- Slope heatmap
- Buildable zone proposal
- No-build or high-risk zone proposal
- Suggested house placement zones ranked by practicality

## 12. Zoning and Compliance Engine

This engine should not only read planning documents. It must convert legal wording into actionable design rules.

### 12.1 Rules to extract where applicable

- Zoning category and use rights
- Density rules
- Coverage and FAR
- Street, side and rear building lines
- Height restrictions
- Parking requirements
- Second dwelling, subdivision, consent use or additional rights
- Servitudes and overlays
- Special conditions tied to the property

### 12.2 Practical outputs



- Buildable envelope overlay
- Warnings where the concept exceeds rules
- Suggestion whether a double-storey concept is more efficient than single-storey
- Suggestion whether a stand is too constrained for certain house types
- Flagging of uncertainty where title conditions or estate rules may conflict with zoning

## 13. Climate, Sun, Wind and Natural Elements Engine

This module matters because real house comfort, outdoor usability, solar value, and even the feel of the home depend on climate and natural conditions.

### 13.1 Factors to analyse

- Seasonal sun path and roof exposure
- Prevailing wind directions and stronger gust exposures
- Local rainfall pattern and water harvesting opportunity
- Stormwater flow tendencies on the stand
- Heat gain risk on west façades
- Need for shading, pergolas, overhangs or smaller openings on harsh sides
- Outdoor comfort zones for a braai, patio, pool or courtyard
- Likely best positions for water tanks

### 13.2 Design influence

- Living areas should receive the best sunlight that suits the climate and user priorities.
- Outdoor entertainment zones should be placed where wind and sun are comfortable, not merely where there is leftover space.
- Water tanks, service yards, and less glamorous functions should be placed where they do not damage the main architecture or key outdoor zones.
- Solar panels and geyser options should be linked to the roof geometry produced by the house-layout engine.

## 14. Look, Feel and Visual Style Engine

The platform must allow the user to guide the architectural language. This is not cosmetic. Style affects massing, roof form, openings, façade rhythm, and the estate's visual identity.

### 14.1 Inputs

- Architectural style family
- Roof preference or estate roof rules
- Material palette
- Desired feeling, for example warm, premium, clean, family-friendly, natural, or minimal
- Estate house-type family where applicable

### 14.2 Outputs

- Exterior render options
- Facade variations
- Material swap previews
- Streetscape consistency logic for estate projects

## 15. Estate-Level Intelligence

For a development like The Greens, the platform should first analyse the whole estate, then allow stand-by-stand design within that estate logic. It must use one core engine operating at two levels, not two separate disconnected systems.

### 15.1 Estate-level analysis

- Read the SDP and stand boundaries
- Understand roads, open spaces, entrances and circulation
- Classify stands by type, such as corner, narrow, premium, sloped, exposed or privacy-sensitive
- Identify which stands are best suited to standard house types and which should be more tailored
- Recommend entrance and security strategies
- Consider streetscape quality, façade rhythm and repetition efficiency
- Review estate-wide sustainability opportunities

### 15.2 Stand-by-stand estate design

- Each stand should inherit estate rules, style limits, neighbouring context, and road context.
- The user should be able to select a standard house type and adapt it, or generate a fully stand-specific design.
- The platform should know whether the stand is premium, constrained, slope-sensitive, or privacy-sensitive before designing the house.

## 16. Security and Entrance Planning Engine

Estate projects should include a dedicated security planning layer, because access control and entrance design are part of development practicality, resident experience, and sales value.

| Decision area          | Examples the platform should compare                                                            |
|------------------------|-------------------------------------------------------------------------------------------------|
| Resident access        | Facial recognition, ANPR, app-based access, remotes, tags, biometrics                           |
| Visitor access         | QR visitor codes, app invitations, intercom and approval flow                                   |
| Perimeter security     | CCTV, electric fence, beams, thermal or standard camera layers                                  |
| Operational resilience | Power backup, communication backup, offline fail-safe logic                                     |
| Guardhouse design      | Single lane vs dual lane, inbound/outbound flow, pedestrian gate, staff toilet and office needs |

## 17. Green and Efficiency Recommendations

The platform should not present green solutions as generic extras. It should match them to climate, stand geometry, roof form, estate logic, and budget level.

- Solar PV positioning and approximate roof suitability
- Solar or heat-pump geyser recommendations
- Rainwater harvesting logic and tank placement
- Greywater or irrigation concepts where appropriate
- Passive shading and cross-ventilation suggestions
- Material choices that reduce maintenance or heat gain
- Landscape ideas that reduce water use

- Value-engineering comparisons between greener and standard options

## **18. Structural and Buildability Engine**

Every concept should be checked for build practicality. The platform should prefer elegant but rational construction over complex geometry unless the user explicitly asks for something more ambitious.

- Foundation type suggestions based on slope and site conditions
- Retaining wall risk or likely need
- Simpler roof and span logic where possible
- Stacking of bathrooms and wet services for efficiency
- Assessment of whether the plan is too fragmented or over-complicated
- Construction complexity score

## **19. Cost and Bill of Quantities Engine**

Once a concept is selected, the system should produce a developer-friendly cost picture and a preliminary quantity structure.

### **19.1 Cost outputs**

- Estimated build cost range
- Estimated rate per square metre
- Earthworks and retaining sensitivity note
- Main structure and finishes split
- External works allowances
- Comparison between options and cost-saving alternatives

### **19.2 BOQ outputs**

- Preliminary quantities by trade
- Walling, roof, floor area and openings schedules
- Major external elements such as paving, walls, patios and tanks
- Exportable format for QS or builder refinement

## 20. Functional Screen and Module Layout

| Module            | Main screen purpose                                   | Key outputs                                                |
|-------------------|-------------------------------------------------------|------------------------------------------------------------|
| Project Setup     | Create project and choose mode                        | Project shell and document intake                          |
| Data Room         | Upload land, legal, estate and climate-related inputs | Structured project data library                            |
| Site Intelligence | Visualise terrain, envelope and constraints           | 3D site model, slope map, buildable zones                  |
| Brief Builder     | Capture structured house and style requirements       | Validated design brief                                     |
| Concept Generator | Generate layout options                               | Plans, site placement, massing                             |
| Visuals           | Show renders and style options                        | Exterior views and comparisons                             |
| Climate and Green | Show solar, wind, water and outdoor logic             | Solar placement, water tank, patio and shading suggestions |
| Estate Dashboard  | Analyse full development and stands                   | Stand classes, security logic, estate recommendations      |
| Cost and BOQ      | Summarise cost and quantities                         | Budget view and quantity output                            |
| Export Centre     | Generate reports and packs                            | Concept pack, feasibility summary, PDFs and schedules      |

## 21. Data Architecture and Technical Integrations

### 21.1 Core technical components

| Component                    | Role in platform                                                                              |
|------------------------------|-----------------------------------------------------------------------------------------------|
| Map and geospatial layer     | Store site location, boundaries, overlays and context                                         |
| Terrain engine               | Build terrain model from contours and support slope analysis                                  |
| Document intelligence engine | Parse zoning certificates, title deeds, estate rules and related PDFs                         |
| Rules engine                 | Convert extracted rules into constraints and design checks                                    |
| Layout generation engine     | Create site-aware house plans and level strategies                                            |
| Visual generation layer      | Produce exterior render views and material/style variants                                     |
| Climate engine               | Analyse sun, wind, rainfall and outdoor comfort                                               |
| Cost engine                  | Generate rough cost logic and BOQ structure                                                   |
| Project database             | Store parent estate projects and child stand projects, along with all user inputs and outputs |

### 21.2 Suggested integration categories

- Mapping / geospatial services
- Terrain and contour processing capability
- CAD or geometry conversion layer for survey files
- Document OCR and rule-extraction model
- Generative planning and visualisation engine
- Climate and solar-data feed
- Cost database specific to South African build assumptions
- Export engine for PDF, schedules, and quantity outputs

## 22. Logic for Parent and Child Projects

Estate projects should be structured as a parent project containing all site-wide logic, with each stand operating as a child project that inherits relevant constraints and estate rules.

| Parent estate project contains                                                                       | Child stand project contains                                                                                         |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| SDP, roads, entrance logic, estate guidelines, stand database, security strategy, master constraints | Stand geometry, stand-specific slope, user house brief, stand-specific layout options, renders, cost and BOQ outputs |

## 23. Required Reports and Deliverables

| Report / export              | Purpose                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------|
| Developer feasibility report | Summarises site, rules, risks, concept options and recommendation                                        |
| Stand concept pack           | Shows plan, site placement, exterior renders, design rationale and cost summary                          |
| Estate strategy report       | Summarises stand classification, security, entrances, house-type strategy and estate-level opportunities |
| Green and climate summary    | Explains solar, water, shading and outdoor-living recommendations                                        |
| Preliminary BOQ              | Supports builder or QS review                                                                            |

## 24. Recommended MVP Scope

The platform should not attempt to build every advanced feature at once. The first release should prove the layout engine and show that the tool can convert real stand data into useful concepts.

| Priority  | Include in MVP                                                                                                                                                          | Hold for later                                                                                                                              |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Must have | Stand input, contour upload, zoning upload, buildable envelope, structured house brief, 2-3 site-aware layouts, exterior renders, basic solar note, rough cost estimate |                                                                                                                                             |
| Phase 2   |                                                                                                                                                                         | More advanced climate engine, estate masterplan mode, security engine, stand categorisation, stronger BOQ logic                             |
| Phase 3   |                                                                                                                                                                         | Full highest-and-best-use engine, deeper services integration, advanced multi-unit optimisation, more automated municipal readiness scoring |

## 25. Quality Standards for the House-Layout Engine

Because the house-layout engine is central to the platform, every generated option should be judged against a quality checklist.

- Does the plan clearly respond to the stand shape and contours?
- Are levels used intelligently and economically?
- Is the garage approach practical from the street?
- Are living areas positioned in the best sun zone for the context?
- Are outdoor entertainment spaces comfortable in terms of wind and sun?
- Is privacy from neighbours handled properly?
- Are passages and wasted circulation kept under control?
- Is the concept visually strong enough to support estate quality and sales appeal?
- Would a builder regard the concept as sensible to construct?
- Does the exterior look and feel match the intended product level and estate character?

## 26. Final Recommended Platform Definition

The platform should be defined as an AI-powered development and design system that turns South African land data, planning rules, contours, climate, and user requirements into practical house layouts, estate intelligence, exterior visualisation, build-efficiency guidance, sustainability recommendations, and preliminary quantity outputs.

### **Bottom line**

The platform succeeds or fails on whether it can produce excellent stand-specific house layouts and convincing look-and-feel outputs. Every other engine should support that goal.

## 27. Immediate Next Steps After Approval of This Scope

16. Lock the product scope and module list.
17. Define detailed field-by-field requirements for the Brief Builder.
18. Map the user journey for Single Erf, Estate Masterplan, and Estate Stand modes.
19. Design the screen architecture and dashboard structure.
20. Prioritise the minimum technical integrations needed to prove the concept.
21. Prototype the house-layout workflow first before expanding to broader estate and BOQ layers.